ECON 0100 | Exercise D1 | Game Theory

Q1 | Study Room Dynamics

Students in Ravenclaw and Hufflepuff share a secondary study room, which is open to all students from both houses at all hours. Every year the heads of house from both houses organize the sharing of cleaning duties. Cleaning improves the quality of the study room for members of both houses, but the chore is inconvenient. If both houses clean, the study room is a fantastic resource. But cleaning takes precious time away from studying, so neither house prefers doing it. The following matrix represents the benefits of each possible strategy combination.

		Hufflepuff	
		Clean	Don't Clean
Ravenclaw	Clean	10, 10	6, 12
	Don't Clean	12, 6	8, 8

a) What is Ravenclaw's best response?

b) What is Hufflepuff's best response?

c) Identify each house's dominant strategy if one exists.

d) Find all Nash Equilibrium.

e) Find the socially efficient strategies.

Q2 | Date Night

Hagrid and McGonagall planned a date night, but had forgotten to decide between two locations: the Leaky Cauldron or Hog's Head Pub. The time for the date was quickly approaching but neither knew where the other was going and didn't have an owl with which to communicate. McGonagall knew Hagrid likes going out with McGonagall ($\Pi = 1$) more than going out alone ($\Pi = 0$), but that he likes going out with her to Hog's head the best ($\Pi = 2$). Hagrid knew McGonagall likes going out to Hog's Head equally whether with or without Hagrid ($\Pi = 1$), but that she only likes going to the Leaky Cauldron alone ($\Pi = 0{,}1$).

a) First, use the information about their preferences to setup the payoff matrix.

		McGonagall	
		Leaky Cauldron	Hog's Head
Hagrid	Leaky Cauldron		
	Hog's Head		

b) Second, use the boxing method to find both players best responses above.

c) Third, find the Nash Equilibrium. Since they couldn't communicate, could they use this to figure out which location to go?